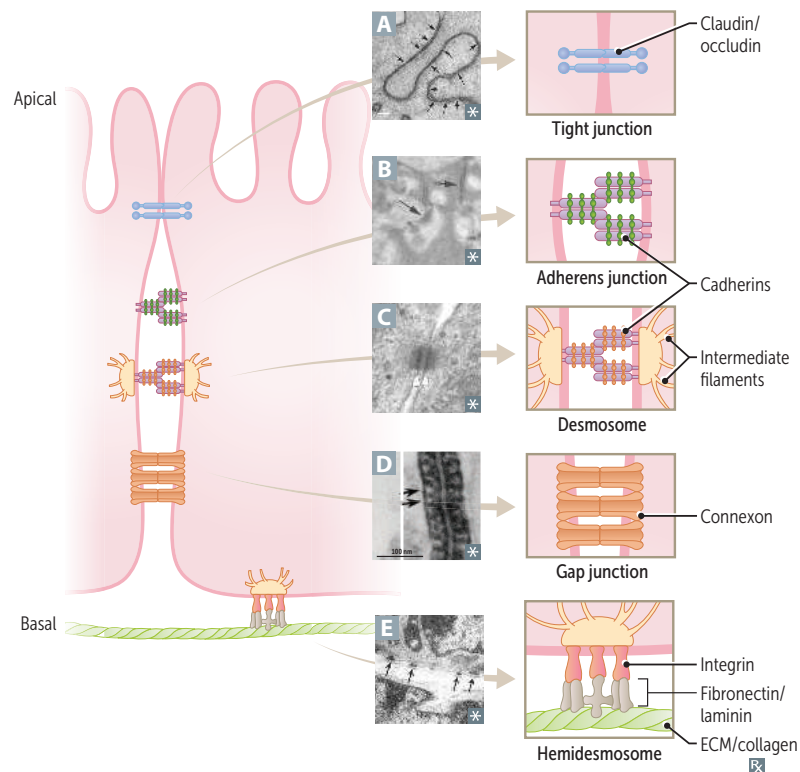


Epithelial cell junctions



Tight junctions (zonula occludens) **A**—prevents paracellular movement of solutes; composed of claudins and occludins.

Adherens junction (belt desmosome, zonula adherens) **B**—forms “belt” connecting actin cytoskeletons of adjacent cells with **cadherins** (Ca^{2+} -dependent **adhesion** proteins). Loss of E-cadherin promotes metastasis.

Desmosome (spot desmosome, macula adherens) **C**—structural support via intermediate filament interactions. Autoantibodies to desmoglein 3 +/- desmoglein 1 → pemphigus vulgaris.

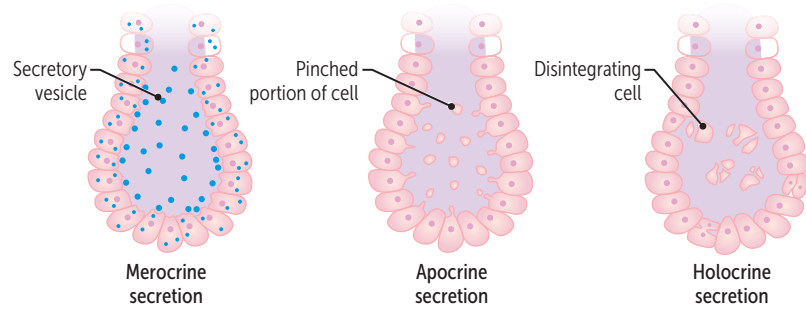
Gap junction **D**—channel proteins called connexons permit electrical and chemical communication between cells.

Hemidesmosome **E**—connects keratin in basal cells to underlying basement membrane. Autoantibodies → **bullous** pemphigoid. (Hemidesmosomes are down “**bullo**.”)

Integrins—membrane proteins that maintain **integrity** of basolateral membrane by binding to collagen, laminin, and fibronectin in basement membrane.

Exocrine glands

Glands that produce substances other than hormones (vs endocrine glands, which secrete hormones) that are released through ducts to the exterior of the body. Can be merocrine (eg, salivary and sweat glands), apocrine (eg, mammary glands), or holocrine (eg, sebaceous glands).



Dermatologic macroscopic terms

LESION	CHARACTERISTICS	EXAMPLES
Macule	Flat lesion with well-circumscribed change in skin color < 1 cm	Freckle (ephelis), labial macule A
Patch	Macule > 1 cm	Vitiligo B
Papule	Elevated solid skin lesion < 1 cm	Neurofibroma C , acne
Plaque	Papule > 1 cm	Psoriasis D
Vesicle	Small fluid-containing blister < 1 cm	Chickenpox (varicella), shingles (zoster) E
Bulla	Large fluid-containing blister > 1 cm	Bullous pemphigoid F
Pustule	Vesicle containing pus	Pustular psoriasis G
Wheal	Transient smooth papule or plaque	Hives (urticaria) H
Scale	Flaking off of stratum corneum	Eczema, psoriasis, SCC I
Crust	Dry exudate	Impetigo J
Ulcer	Epidermal loss leading to exposure of the basement membrane and underlying structures (ie, dermis, muscle, bone, or tendon)	Arterial ulcer, decubitus ulcer, leprosy
Erosion	Partial or full loss of epidermis without exposure of the basement membrane	Can be seen in GERD; erosive esophagitis

**Dermatologic microscopic terms**

LESION	CHARACTERISTICS	EXAMPLES
Dyskeratosis	Abnormal premature keratinization	Squamous cell carcinoma
Hyperkeratosis	↑ thickness of stratum corneum	Psoriasis, calluses
Parakeratosis	Retention of nuclei in stratum corneum	Psoriasis, actinic keratosis
Hypergranulosis	↑ thickness of stratum granulosum	Lichen planus
Spongiosis	Epidermal accumulation of edematous fluid in intercellular spaces	Eczematous dermatitis
Acantholysis	Separation of epidermal cells	Pemphigus vulgaris
Acanthosis	Epidermal hyperplasia (↑ spinosum)	Acanthosis nigricans, psoriasis

Pigmented skin disorders

Albinism	Normal melanocyte number with ↓ melanin production A due to ↓ tyrosinase activity or defective tyrosine transport. ↑ risk of skin cancer.
Melasma (chloasma)	Acquired hyperpigmentation associated with pregnancy (“mask of pregnancy” B) or OCP use. More common in patients with darker skin tones.
Vitiligo	Irregular patches of complete depigmentation C . Caused by destruction of melanocytes (believed to be autoimmune). Associated with other autoimmune disorders.
Waardenburg syndrome	Patchy depigmentation of skin, hair, and irises that can be associated with deafness. Caused by defects in the differentiation of neural crest cells into melanocytes.

**Seborrheic dermatitis**

Erythematous, well-demarcated plaques **A** with greasy yellow scales in areas rich in sebaceous glands, such as scalp, face, and periocular region. Common in both infants (cradle cap) and adults. Extensive disease may be associated with HIV infection and Parkinson disease. Sebaceous glands are not inflamed, but play a role in disease development. Possibly associated with *Malassezia* spp. Treatment: topical antifungals and glucocorticoids.

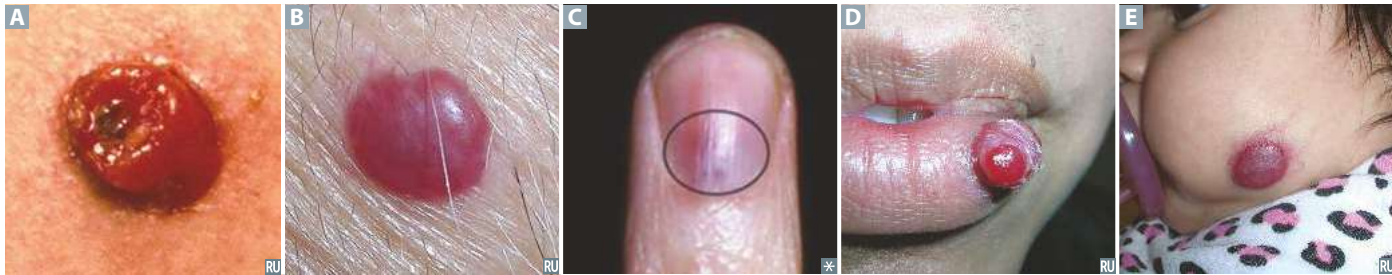
Common skin disorders

Acne	Multifactorial etiology—↑ sebum/androgen production, abnormal keratinocyte desquamation, <i>Cutibacterium acnes</i> colonization of the pilosebaceous unit (comedones), and inflammation (papules/pustules A , nodules, cysts). Treatment: retinoids, benzoyl peroxide, and antibiotics.
Atopic dermatitis (eczema)	Pruritic eruption associated with ichthyosis vulgaris and other atopic diseases (asthma, allergic rhinitis, food allergies); ↑ serum IgE. Often appears on face in infancy B and then on flexural surfaces C in children and adults.
Allergic contact dermatitis	Type IV hypersensitivity reaction secondary to contact allergen (eg, nickel D , poison ivy E , neomycin).
Keratosis pilaris	Follicular-based papules F from keratin plugging, most often on extensor surfaces of arms and thighs.
Melanocytic nevus	Common mole. Benign, but melanoma can arise in congenital or atypical moles. Intradermal nevi are papular G . Junctional nevi are flat macules H .
Pseudofolliculitis barbae	Inflammatory reaction to hair penetrating the skin characterized by firm papules and pustules that are painful and pruritic. Commonly occurs near jawline as a result of shaving (“razor bumps”), more common with naturally curly hair.
Psoriasis	Papules and plaques with silvery scaling, especially on knees and elbows. Acanthosis with parakeratotic scaling (nuclei still in stratum corneum), Munro microabscesses. ↑ stratum spinosum, ↓ stratum granulosum. Auspitz sign (I)—pinpoint bleeding spots from exposure of dermal papillae when scales are scraped off. Associated with nail pitting and psoriatic arthritis.
Rosacea	Inflammatory facial skin disorder characterized by erythematous papules and pustules J , but no comedones. May be associated with facial flushing in response to external stimuli (eg, alcohol, heat). Complications include ocular involvement, rhinophyma (bulbous deformation of nose).
Seborrheic keratosis	Well-demarcated, verrucous, benign squamous epithelial proliferation of immature keratinocytes with keratin-filled cysts (horn cysts) K . Looks “stuck on.” Leser-Trélat sign L —rapid onset of multiple seborrheic keratoses, indicates possible malignancy (eg, GI adenocarcinoma).
Verrucae	Warts; caused by low-risk HPV strains. Soft, tan-colored, cauliflowerlike papules M . Epidermal hyperplasia, hyperkeratosis, koilocytosis. Condyloma acuminatum on anus or genitals N .
Urticaria	Hives. Pruritic wheals that form after mast cell degranulation O . Characterized by superficial dermal edema and lymphatic channel dilation.



Vascular tumors of skin

Angiosarcoma	Rare blood vessel malignancy typically occurring in the head, neck, and breast areas. Usually in older adults, on sun-exposed areas. Associated with radiation therapy and chronic postmastectomy lymphedema. Stewart-Treves syndrome —cutaneous angiosarcoma developing after chronic lymphedema. Hepatic angiosarcoma associated with vinyl chloride and arsenic exposures. Very aggressive and difficult to resect due to delay in diagnosis.
Bacillary angiomatosis	Benign capillary skin papules A found in patients with AIDS. Caused by <i>Bartonella</i> infections. Frequently mistaken for Kaposi sarcoma, but has neutrophilic infiltrate.
Cherry angioma	Benign capillary hemangioma B commonly appearing in middle-aged adults. Does not regress. Frequency ↑ with age.
Glomus tumor	Benign, painful, red-blue tumor, commonly under fingernails C . Arises from modified smooth muscle cells of the thermoregulatory glomus body.
Kaposi sarcoma	Endothelial malignancy most commonly affecting the skin, mouth, GI tract, respiratory tract. Classically seen in older Eastern European males, patients with AIDS, and organ transplant patients. Associated with HHV-8 and HIV. Lymphocytic infiltrate, unlike bacillary angiomatosis.
Pyogenic granuloma	Polypoid lobulated capillary hemangioma D that can ulcerate and bleed. Associated with trauma and pregnancy.
Infantile hemangioma	Benign capillary hemangioma of infancy E . Appears in first few weeks of life (1/200 births); initially grows rapidly, then involutes starting at age 1. I nfantile hemangiomas spontaneously i nvolute; c herry angiomas c annot.



Skin infections**Bacterial infections**

Impetigo	Skin infection involving superficial epidermis. Usually from <i>S aureus</i> or <i>S pyogenes</i> . Highly contagious. Honey-colored crusting A . Bullous impetigo B has bullae and is usually caused by <i>S aureus</i> .
Erysipelas	Infection involving upper dermis and superficial lymphatics, usually from <i>S pyogenes</i> . Presents with well-defined, raised demarcation between infected and normal skin C .
Cellulitis	Acute, painful, spreading infection of deeper dermis and subcutaneous tissues. Usually from <i>S pyogenes</i> or <i>S aureus</i> . Often starts with a break in skin from trauma or another infection D .
Abscess	Collection of pus from a walled-off infection within deeper layers of skin E . Offending organism is almost always <i>S aureus</i> .
Necrotizing fasciitis	Deeper tissue injury, usually from anaerobic bacteria or <i>S pyogenes</i> . Pain may be out of proportion to exam findings. Results in crepitus from methane and CO ₂ production. “Flesh-eating bacteria.” Causes bullae and skin necrosis → violaceous color of bullae, surrounding skin F . Surgical emergency.
Staphylococcal scalded skin syndrome	Exotoxin destroys keratinocyte attachments in stratum granulosum only (vs toxic epidermal necrolysis, which destroys epidermal-dermal junction). No mucosal involvement. Characterized by fever and generalized erythematous rash with sloughing of the upper layers of the epidermis G that heals completely. ⊕ Nikolsky sign (separation of epidermis upon manual stroking of skin). Commonly seen in newborns and children/adults with renal insufficiency.

Viral infections

Herpes	Herpes virus infections (HSV-1 and HSV-2) of skin can occur anywhere from mucosal surfaces to normal skin. These include herpes labialis, herpes genitalis, herpetic whitlow H (finger).
Molluscum contagiosum	Umbilicated papules I caused by a poxvirus. While frequently seen in children, it may be sexually transmitted in adults.
Varicella zoster	Causes varicella (chickenpox) and zoster (shingles). Varicella presents with multiple crops of lesions in various stages from vesicles to crusts. Zoster is a reactivation of the virus in dermatomal distribution (unless it is disseminated).
Hairy leukoplakia	Irregular, white, painless plaques on lateral tongue that cannot be scraped off J . EBV mediated. Occurs in patients living with HIV, organ transplant recipients. Contrast with thrush (scrapable) and leukoplakia (precancerous).



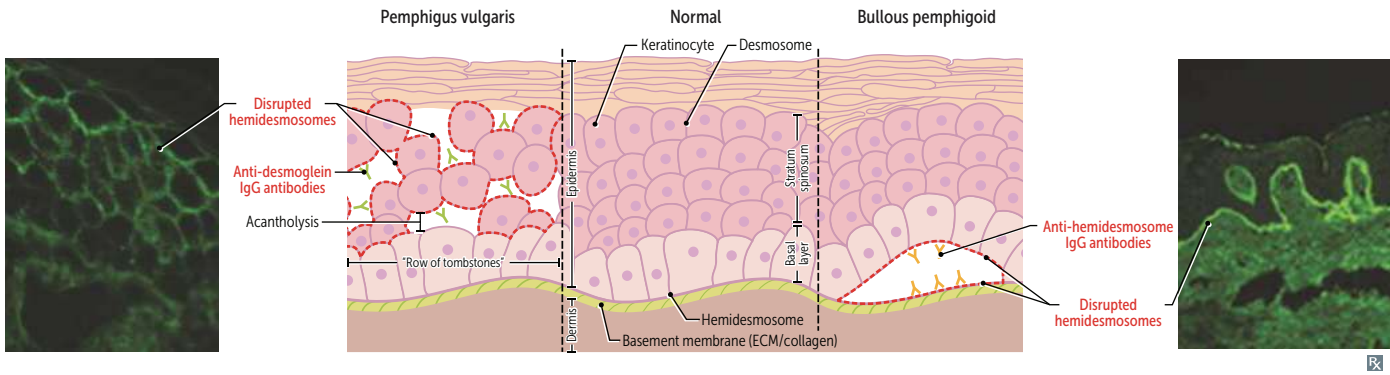
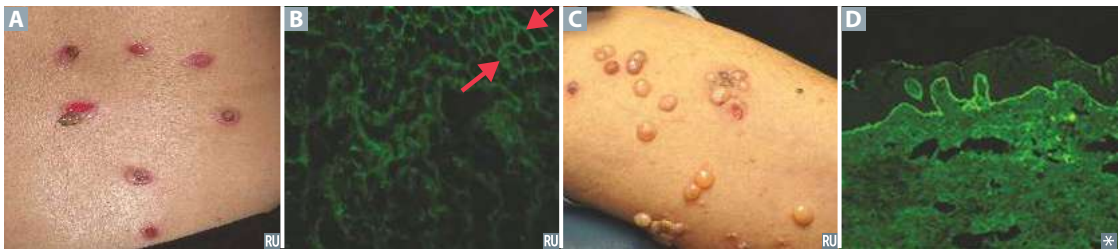
Cutaneous mycoses

Tinea (dermatophytes)	Clinical name for dermatophyte (cutaneous fungal) infections. Dermatophytes include <i>Microsporum</i> , <i>Trichophyton</i> , and <i>Epidermophyton</i> . Branching septate hyphae visible on KOH preparation with blue fungal stain A . Associated with pruritus.
Tinea capitis	Occurs on head, scalp. Associated with lymphadenopathy, alopecia, scaling B .
Tinea corporis	Occurs on body (usually torso). Characterized by enlarging erythematous, scaly rings (“ringworm”) with central clearing C . Can be acquired from contact with infected pets or farm animals.
Tinea cruris	Occurs in inguinal area (“jock itch”) D . Often does not show the central clearing seen in tinea corporis.
Tinea pedis	Three varieties (“athlete’s foot”): ▪ Interdigital E; most common ▪ Moccasin distribution F ▪ Vesicular type
Tinea unguium	Onychomycosis; occurs on nails.
Tinea (pityriasis) versicolor	Caused by <i>Malassezia</i> spp. (<i>Pityrosporum</i> spp.), a yeastlike fungus (not a dermatophyte despite being called tinea). Degradation of lipids produces acids that inhibit tyrosinase (involved in melanin synthesis) → hypopigmentation G ; hyperpigmentation and/or pink patches can also occur due to inflammatory response. Less pruritic than dermatophytes. Can occur any time of year, but more common in summer (hot, humid weather). “Spaghetti and meatballs” appearance on microscopy H . Treatment: selenium sulfide, topical and/or oral antifungal medications.



Autoimmune blistering skin disorders

	Pemphigus vulgaris	Bullous pemphigoid
PATHOPHYSIOLOGY	Potentially fatal. Most commonly seen in older adults. Type II hypersensitivity reaction. IgG antibodies against desmoglein 3 +/- desmoglein 1 (component of desmosomes, which connect keratinocytes in the stratum spinosum).	Less severe than pemphigus vulgaris. Most commonly seen in older adults. Type II hypersensitivity reaction. IgG antibodies against hemidesmosomes (epidermal basement membrane; antibodies are “bullo” the epidermis).
GROSS MORPHOLOGY	Flaccid intraepidermal bullae A caused by acantholysis (separation of keratinocytes, “row of tombstones” on H&E stain); oral mucosa is involved. Nikolsky sign ⊕.	Tense blisters C containing eosinophils; oral mucosa spared. Nikolsky sign ⊖.
IMMUNOFLUORESCENCE	Reticular pattern around epidermal cells B .	Linear pattern at epidermal-dermal junction D .



Other blistering skin disorders**Dermatitis
herpetiformis**

Pruritic papules, vesicles, and bullae (often found on elbows, knees, buttocks) **A**. Deposits of IgA at tips of dermal papillae. Associated with celiac disease. Treatment: dapsone, gluten-free diet.

Erythema multiforme

Associated with infections (eg, *Mycoplasma pneumoniae*, HSV), drugs (eg, sulfa drugs, β -lactams, phenytoin). Presents with multiple types of lesions—macules, papules, vesicles, target lesions (look like targets with multiple rings and dusky center showing epithelial disruption) **B**.

**Stevens-Johnson
syndrome**

Characterized by fever, bullae formation and necrosis, sloughing of skin at dermal-epidermal junction ($\oplus$ Nikolsky), high mortality rate. Typically mucous membranes are involved **C**. Targetoid skin lesions may appear, as seen in erythema multiforme. Usually associated with adverse drug reaction. **Toxic epidermal necrolysis (TEN)** **D E** is more severe form of SJS involving $> 30\%$ body surface area. 10–30% involvement denotes SJS-TEN.

**Inherited
Epidermolysis
Bullosa**

Group of inherited skin disorders characterized by bullae, erosions, and ulcers triggered by minor trauma. Presents with blisters early in life and oral blisters with bottle feeding. Most commonly caused by mutations in keratin genes.



Cutaneous ulcers

	Venous ulcer	Arterial ulcer	Neuropathic ulcer	Pressure injury
ETIOLOGY	Chronic venous insufficiency; most common ulcer type	Peripheral artery disease (eg, atherosclerotic stenosis)	Peripheral neuropathy (eg, diabetic foot)	Prolonged unrelieved pressure (eg, immobility)
LOCATION	Gaiter area (ankle to midcalf), typically over malleoli	Distal toes, anterior shin, pressure points	Bony prominences (eg, metatarsal heads, heel)	Weightbearing points (eg, sacrum, ischium, calcaneus)
APPEARANCE	Irregular border, shallow, exudative A	Symmetric with well-defined punched-out appearance B	Hyperkeratotic edge with undermined borders C	Varies based on stage from non-blanchable erythema to full-thickness skin loss D
PAIN	Mild to moderate	Severe	Absent	Present
ASSOCIATED SIGNS	Telangiectasias, varicose veins, edema, stasis dermatitis (erythematous eczematous patches)	Arterial insufficiency, cold and pale atrophic skin, hair loss, absent pulses	Claw toes, Charcot joints, absent reflexes	Soft tissue infection and osteomyelitis are frequent complications



Miscellaneous skin disorders

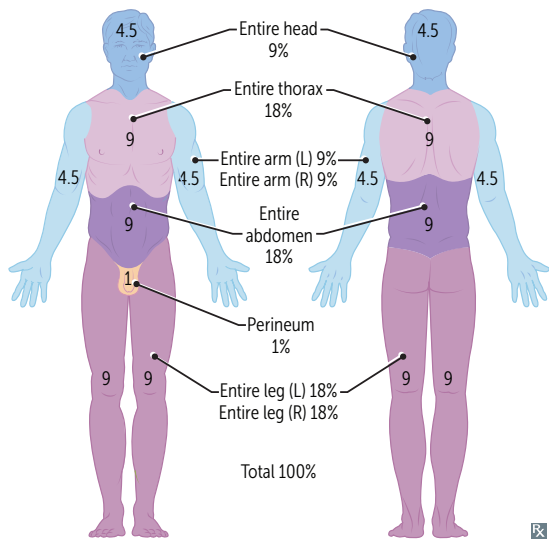
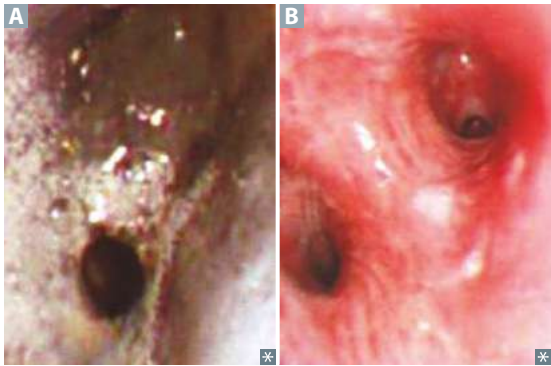
Acanthosis nigricans	Epidermal hyperplasia causing symmetric, hyperpigmented thickening of skin, especially in axilla or on neck A . Associated with insulin resistance (eg, diabetes, obesity, Cushing syndrome, PCOS), visceral malignancy (eg, gastric adenocarcinoma).
Erythema nodosum	Painful, raised inflammatory lesions of subcutaneous fat (panniculitis), usually on anterior shins. Often idiopathic, but can be associated with sarcoidosis, coccidioidomycosis, histoplasmosis, TB, streptococcal infections B , leprosy C , inflammatory bowel disease.
Ichthyosis vulgaris	Disorder of defective keratinocyte desquamation due to filaggrin gene mutations resulting in diffuse scaling of the skin D most commonly on the extensor side of extremities and the trunk. Manifests in infancy or early childhood. Strong association with atopic dermatitis.
Lichen Planus	Pruritic, purple, polygonal planar papules and plaques are the 6 P's of lichen Planus E F . Mucosal involvement manifests as Wickham striae (reticular white lines) and hypergranulosis. Sawtooth infiltrate of lymphocytes at dermal-epidermal junction. Associated with hepatitis C.
Pityriasis rosea	“Herald patch” G followed days later by other scaly erythematous plaques, often in a “Christmas tree” distribution on trunk H . Multiple pink plaques with collarette scale. Self-resolving in 6–8 weeks.
Sunburn	Acute cutaneous inflammatory reaction due to excessive UV irradiation. Causes DNA mutations, inducing apoptosis of keratinocytes. UVB is dominant in sunBurn, UVA in tAnning and photoAging. Exposure to UVA and UVB ↑ risk of skin cancer.
Radiation dermatitis	Can be acute or late. Acute occurs ≤ 90 days after radiotherapy due to apoptosis of basal keratinocytes and epidermal edema. Presents with erythema, desquamation, superficial ulceration, and blistering. Late occurs months to years after radiotherapy due to fibrosis with homogenization of dermal collagen fibers on histology, vascular damage, and telangiectasias.



Estimation of body surface area

Approximated by the rule of 9's. Used to assess the extent of burn injuries.

Inhalation injury—complication of inhalation of noxious stimuli (eg, smoke). Heat, particulates (< 1 μm diameter), or irritants (eg, NH₃) → chemical tracheobronchitis, edema, pneumonia, acute respiratory distress syndrome. Singed nasal hairs or soot in oropharynx common on exam. Bronchoscopy shows severe edema, congestion of bronchus, and soot deposition (**A**, 18 hours after inhalation injury; **B**, resolution at 11 days after injury).



Burn classification

DEPTH	INVOLVEMENT	APPEARANCE	SENSATION
Superficial burn	Epidermis only	Similar to sunburn; histamine release causes localized, dry, blanching redness without blisters	Painful
Superficial partial-thickness burn	Epidermis and papillary dermis	Blisters, blanches with pressure, swollen, warm	Painful to temperature and air
Deep partial-thickness burn	Epidermis and reticular dermis	Blisters (easily unroofed), does not blanch with pressure	Painless; perception of pressure only
Full-thickness burn	Epidermis and full-thickness dermis	White, waxy, dry, inelastic, leathery, does not blanch with pressure	Painless; perception of deep pressure only
Deeper injury burn	Epidermis, dermis, and involvement of underlying tissue (eg, fascia, muscle)	White, dry, inelastic, does not blanch with pressure	Painless; some perception of deep pressure

Skin cancer

Basal cell carcinoma (BCC) more common above **upper lip**.

Squamous cell carcinoma (SCC) more common below **lower lip**.

Sun exposure strongly predisposes to skin cancer.

**Basal cell carcinoma**

Most common skin cancer. Found in sun-exposed areas of body (eg, face). Locally invasive, but rarely metastasizes. Waxy, pink, pearly nodules, commonly with telangiectasias, rolled borders **A**, central crusting or ulceration. BCCs also appear as a scaling plaque (superficial BCC) **B**.

Squamous cell carcinoma

Second most common skin cancer. Associated with immunosuppression, chronic nonhealing wounds, and occasionally arsenic exposure. **Marjolin ulcer**—SCC arising in chronic wounds or scars; usually develops > 20 years after insult. Commonly appears on face **C**, lower lip **D**, ears, hands. Locally invasive, may spread to lymph nodes, and will rarely metastasize. Ulcerative red lesions. Histopathology: keratin “pearls” **E**.

Actinic keratosis—Premalignant lesions caused by sun exposure. Small, rough, erythematous or brownish papules or plaques **F**. Risk of squamous cell carcinoma is proportional to degree of epithelial dysplasia.

Melanoma

Common tumor with significant risk of metastasis. S-100 tumor marker. Associated with dysplastic nevi; people with lighter skin tones are at ↑ risk. Depth of tumor (Breslow thickness) correlates with risk of metastasis. Look for the **ABCDEs**: **A**symmetry, **B**order irregularity, **C**olor variation, **D**iameter > 6 mm, and **E**volution over time. At least 4 different types of melanoma, including superficial spreading **G**, nodular **H**, lentigo maligna **I**, and acral lentiginous (highest prevalence in people with darker skin tones) **J**. Often driven by activating mutation in BRAF kinase.

Primary treatment is excision with appropriately wide margins. Advanced melanoma also treated with immunotherapy (eg, ipilimumab) and/or BRAF inhibitors (eg, vemurafenib).

